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👤 DEV 1 — PLAN (AI/Perception: SynapseSignal Perception Layer)

🔷 ROLE

Build the **Perception Layer** of SynapseSignal:

- Detect vehicles
- Identify emergency vehicles
- Classify emergency types
- Provide clean, structured outputs for downstream modules

Dev 1 ensures:

- High detection accuracy
- Clean bounding boxes
- Reliable emergency identification
- Consistent output schema

🔷 PHASED EXECUTION PLAN

🔹 PHASE 1 — DATASET VALIDATION & PREPARATION

Goal: Ensure dataset is clean and usable

Tasks:

- Validate YOLO format
- Check:
  - missing labels
  - invalid annotations
- Ensure class mapping:
  - vehicle
  - emergency_vehicle
- Visualize dataset

- Clean incorrect labels

Deliverable:
✔ Clean, validated dataset

🔹 PHASE 2 — EMERGENCY DATA ENHANCEMENT

Goal: Strengthen emergency vehicle representation

Tasks:

- Ensure presence of:
  - ambulance
  - police
  - fire truck
- Balance dataset
- Augment emergency samples

Deliverable:
✔ Balanced dataset with strong emergency coverage

🔹 PHASE 3 — MODEL 1 (YOLOv8 DETECTOR) TRAINING

Goal: Detect vehicles and emergency vehicles

Tasks:

- Train YOLOv8:
  - classes: vehicle, emergency_vehicle
- Tune:
  - epochs
  - batch size
- Validate:
  - mAP
  - precision/recall

Deliverable:
✔ Trained detection model

🔹 PHASE 4 — MODEL 2 (EMERGENCY CLASSIFIER) DATASET

Goal: Prepare dataset for classification

Tasks:

- Extract cropped images from:
  - emergency_vehicle detections
- Label:
  - ambulance
  - police
  - fire_truck

Deliverable:
✔ Classification dataset

🔹 PHASE 5 — MODEL 2 TRAINING (CLASSIFIER)

Goal: Classify emergency vehicle types

Tasks:

- Train classifier (CNN / lightweight model)
- Evaluate:
  - accuracy
  - confusion matrix

Deliverable:
✔ Trained classifier model

🔹 PHASE 6 — PIPELINE INTEGRATION (MODEL 1 + MODEL 2)

Goal: Build 2-stage perception pipeline

Tasks:

- Run YOLO detection
- Filter emergency_vehicle
- Crop bounding boxes
- Pass to classifier
- Merge outputs

Deliverable:
✔ Integrated perception pipeline

🔹 PHASE 7 — OUTPUT STANDARDIZATION

Goal: Provide structured output for Dev 2

Tasks:

- Define output schema:

```json
{
  "frame_id": "...",
  "objects": [
    {
      "id": "...",
      "class": "vehicle/emergency_vehicle",
      "type": "ambulance/police/fire/null",
      "bbox": [...],
      "confidence": ...
    }
  ]
}
```

Deliverable:
✔ Standardized detection output

🔹 PHASE 8 — PERFORMANCE OPTIMIZATION

Goal: Ensure real-time capability

Tasks:

- Optimize:
  - model size (YOLOv8n/s)
  - inference speed
- Reduce latency

Deliverable:
✔ Real-time inference pipeline

🔹 PHASE 9 — EDGE CASE HANDLING

Goal: Improve robustness

Tasks:

- Handle:
  - occlusion
  - low light
  - dense traffic
- Improve detection reliability

Deliverable:
✔ Robust perception system

🔹 PHASE 10 — VALIDATION & HANDOFF

Goal: Ensure readiness for Dev 2

Tasks:

- Validate:
  - detection accuracy
  - emergency classification accuracy
- Test:
  - sample videos
- Ensure:
  - stable output format

Deliverable:
✔ Production-ready perception module

🔷 DEV 1 SUCCESS CRITERIA

- Accurate vehicle detection
- Reliable emergency detection
- Correct emergency type classification
- Real-time inference achieved
- Clean and consistent output schema
- Robust performance under real conditions

✔ Output ready for **Dev 2 (Traffic Intelligence Layer)**

👤 DEV 1 — MASTER PROMPTS (Jupyter Notebook + Dataset Validation | SynapseSignal)

1️⃣ MASTER PROMPT — COMPLETE DATASET VALIDATION NOTEBOOK

You are a computer vision engineer.

Task:
Generate a **complete Jupyter Notebook** to validate a YOLO dataset for SynapseSignal.

You must include:

1. Dataset Structure Check

- Verify:
  - images/ and labels/ folders
  - 1:1 mapping
- Detect:
  - missing labels
  - orphan labels

---

2. Label Format Validation

- Parse each `.txt` file
- Validate:
  - 5 values per line
  - normalized coordinates (0–1)
- Detect:
  - invalid class IDs
  - malformed rows

---

3. Class Distribution Analysis

- Count:

    - vehicle
    - emergency_vehicle
- Plot distribution (matplotlib)

---

4. Bounding Box Visualization

- Randomly sample images
- Draw bounding boxes
- Display class labels

---

5. Error Reporting

- Print:
    - missing files
    - invalid labels
    - summary stats

Output Format:

- Clean notebook cells
- Modular Python functions
- No pseudo-code

Constraints:

- Must run without modification
- No hardcoded paths
- Must handle large datasets

2️⃣ MASTER PROMPT — DATASET VISUALIZATION NOTEBOOK

You are generating a dataset inspection notebook.

Task:
Create a Jupyter notebook to visualize YOLO annotations.

You must include:

- Load random images

- Convert YOLO → pixel coordinates
- Draw bounding boxes using OpenCV
- Overlay class names
- Display multiple samples

Output Format:

- Python notebook cells
- Reusable functions

Constraints:

- No placeholder code
- Must work with any dataset

3️⃣ MASTER PROMPT — DATA CLEANING PIPELINE NOTEBOOK

You are building a dataset cleaning system.

Task:
Generate a Jupyter notebook to clean YOLO dataset.

You must:

- Remove:
  - invalid label files
  - malformed rows
- Fix:
  - out-of-bound coordinates
- Log:
  - all changes

Output Format:

- Before vs after stats
- Cleaned dataset output

Constraints:

- No silent deletions

- Must preserve valid data

4️⃣ MASTER PROMPT — CLASS DISTRIBUTION & BALANCING NOTEBOOK

You are analyzing dataset balance.

Task:
Generate notebook to compute and fix imbalance.

You must:

- Count class instances
- Plot distribution
- Suggest:
  - augmentation
  - oversampling

Output Format:

- Python code + plots

Constraints:

- Must handle large datasets
- No hardcoded assumptions

5️⃣ MASTER PROMPT — DATA AUGMENTATION PIPELINE

You are designing augmentation for detection models.

Task:
Generate notebook using Albumentations.

You must include:

- Horizontal flip
- Brightness/contrast
- Blur
- Scaling
- Bounding box-safe transforms

Output Format:

- Python code
- YOLO bbox compatible

Constraints:

- Must preserve label correctness
- No distortion of emergency features

6️⃣ MASTER PROMPT — YOLOv8 TRAINING NOTEBOOK

You are training a detection model.

Task:
Generate full Jupyter notebook for YOLOv8 training.

You must include:

1. Install dependencies
2. Load dataset (data.yaml)
3. Train model:
    - epochs
    - batch size
    - image size
4. Save model
5. Print training results

Output Format:

- Executable notebook cells

Constraints:

- Use Ultralytics API
- No deprecated code

7️⃣ MASTER PROMPT — EMERGENCY CLASSIFIER TRAINING NOTEBOOK

You are training Model 2 (classifier).

Task:
Generate notebook for emergency vehicle classification.

You must include:

- Dataset loading (cropped images)
- Model (CNN / lightweight)
- Training loop
- Validation
- Accuracy + confusion matrix

Output Format:

- PyTorch/TensorFlow notebook

Constraints:

- Must be efficient
- No overcomplex model

8️⃣ MASTER PROMPT — INFERENCE PIPELINE NOTEBOOK

You are building the full perception pipeline.

Task:
Generate notebook combining Model 1 + Model 2.

You must:

- Load YOLO model
- Run detection
- Filter emergency_vehicle
- Crop bounding boxes
- Run classifier
- Merge outputs

Output Format:

```
json
```

```
{
  "class": "...",
  "type": "...",
  "bbox": [...],
  "confidence": ...
}
```

Constraints:

- Must run in real-time
- Clean modular code

9️⃣ MASTER PROMPT — MODEL EVALUATION NOTEBOOK

You are evaluating detection performance.

Task:
Generate evaluation notebook.

You must compute:

- mAP@50
- mAP@50-95
- Precision
- Recall

Output Format:

- Metrics + plots

Constraints:

- Use YOLO built-in evaluation
- No incorrect calculations

🔟 MASTER PROMPT — FAILURE ANALYSIS NOTEBOOK

You are analyzing model weaknesses.

Task:
Identify failure cases.

You must detect:

- False positives
- False negatives
- Misclassification

Output Format:

- Categorized failures
- Root cause analysis
- Suggested fixes

🔷 DEV 1 MASTER PROMPT OUTCOME

- Fully validated dataset
- Clean visualization tools
- Automated cleaning pipeline
- Balanced dataset
- Trained YOLO model
- Trained classifier
- Integrated inference pipeline
- Accurate evaluation metrics
- Identified failure cases

✔ Perception layer becomes **production-ready for SynapseSignal**

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👤 DEV 1 — GUIDED LEARNING (SynapseSignal Perception Layer: Dataset → Dual-Model Pipeline)

🔹 PHASE 1 — DATASET VALIDATION

🧠 What You Are Learning

You are learning **how machines read data**, not how humans see images.

⚙️ What Is Happening

- YOLO reads:
  - numbers (labels)
  - normalized coordinates
- Not images directly
- Every `.txt` file = ground truth

🧪 Guided Prompt

Run Dataset Validation Notebook Prompt

🔍 What to Notice

- Missing label → model assumes background
- Wrong format → training crash
- Invalid class ID → undefined learning

🧱 Mental Model

Dataset = training truth → if truth is wrong, model is wrong

🔹 PHASE 2 — DATASET VISUALIZATION

🧠 What You Are Learning

You are learning **how to "see what the model sees"**.

⚙️ What Is Happening

- Convert normalized → pixel coordinates
- Overlay bounding boxes

🧪 Guided Prompt

Run Visualization Notebook

🔍 What to Notice

- Tight vs loose boxes
- Missing annotations
- Wrong labels

🧱 Mental Model

Visualization = debugging your dataset

🔹 PHASE 3 — DATA CLEANING

🧠 What You Are Learning

You are learning **data quality enforcement**.

⚙️ What Is Happening

- Removing invalid annotations
- Fixing coordinate errors
- Logging all corrections

🧪 Guided Prompt

Run Data Cleaning Notebook

🔍 What to Notice

- Small errors → large training impact
- Clean data improves convergence

🧱 Mental Model

Garbage in → garbage out

🔹 PHASE 4 — CLASS DISTRIBUTION ANALYSIS

🧠 What You Are Learning

You are learning **how bias forms in models**.

⚙️ What Is Happening

- Model sees more vehicles than emergency vehicles
- Learns dominant class

🧪 Guided Prompt

Run Distribution Notebook

🔍 What to Notice

- If emergency <5% → ignored
- Balanced data improves recall

🧱 Mental Model

Frequency = importance for model

🔹 PHASE 5 — DATA AUGMENTATION

🧠 What You Are Learning

You are learning **how to simulate real-world variability**.

⚙️ What Is Happening

- Apply transformations:
  - lighting
  - blur
  - scale
- Increase dataset diversity

🧪 Guided Prompt

Run Augmentation Pipeline

🔍 What to Notice

- Improves robustness
- Prevents overfitting

🧱 Mental Model

Augmentation = artificial experience

🔹 PHASE 6 — MODEL 1 (YOLOv8 DETECTOR) TRAINING

🧠 What You Are Learning

You are learning **object detection fundamentals**.

⚙️ What Is Happening

- Model predicts:
  - bounding box
  - class
- Optimizes:
  - localization + classification

🧪 Guided Prompt

Run YOLO Training Notebook

🔍 What to Notice

- Loss decreasing
- mAP improving
- Overfitting signs

🧱 Mental Model

Detection = where + what

🔹 PHASE 7 — MODEL 2 DATA PREPARATION

🧠 What You Are Learning

You are learning **feature specialization**.

⚙️ What Is Happening

- Crop emergency_vehicle regions
- Create classification dataset

🧪 Guided Prompt

Generate cropped dataset

🔍 What to Notice

- Quality of crops matters
- Clear features improve classification

🧱 Mental Model

Focus on relevant regions only

🔹 PHASE 8 — MODEL 2 (CLASSIFIER) TRAINING

🧠 What You Are Learning

You are learning **fine-grained classification**.

⚙️ What Is Happening

- Model distinguishes:
  - ambulance
  - police
  - fire

🧪 Guided Prompt

Run Classifier Training Notebook

🔍 What to Notice

- Confusion between classes
- Class imbalance issues

🧱 Mental Model

Classification = feature recognition

🔹 PHASE 9 — PIPELINE INTEGRATION (MODEL 1 + MODEL 2)

🧠 What You Are Learning

You are learning **multi-stage ML pipelines**.

⚙️ What Is Happening

- Detect → filter → crop → classify
- Combine outputs

🧪 Guided Prompt

Run Inference Pipeline Notebook

🔍 What to Notice

- Latency increases
- Accuracy improves
- Modular design

🧱 Mental Model

Pipeline = chain of specialized models

🔹 PHASE 10 — EVALUATION & FAILURE ANALYSIS

🧠 What You Are Learning

You are learning **model debugging**.

⚙️ What Is Happening

- Identify:
  - false positives
  - false negatives
- Analyze causes

🧪 Guided Prompt

Run Evaluation + Failure Analysis

🔍 What to Notice

- Occlusion errors
- Low-light failures
- Misclassification

🧱 Mental Model

Errors reveal model limitations

🔷 FINAL LEARNING OUTCOME

You now understand:

- How datasets drive model learning
- How to validate and clean data
- How object detection works
- Why class balance matters
- How augmentation improves robustness
- How to train YOLO effectively
- How to build a dual-model pipeline
- How to evaluate and debug models

🧠 FINAL INSIGHT

> You are not just training a model —
> you are building the **eyes of SynapseSignal**

✔ Ready to move to **Dev 2 (Traffic Intelligence Layer)**